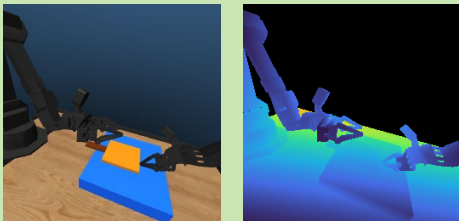
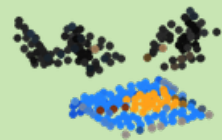


Viewpoint-stable 3D observation

Current RGB-D



Camera-view
PointCloud



Robot-centric
PointCloud



Visual
Encoder



H_t



WorldQueryFormer

state



H_t^F



Fusion



DiT-X



action
chunk

Training-only 4D teacher

Future Clip



t

...



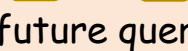
$t + \Delta t$



current query



future query



Z_t^0



Z_t^F



$\mathcal{L}_{cur}$

$\mathcal{L}_{evo}$

Future-Aware Action Policy

Action Execution



arm1



arm0



arm2

Execute

